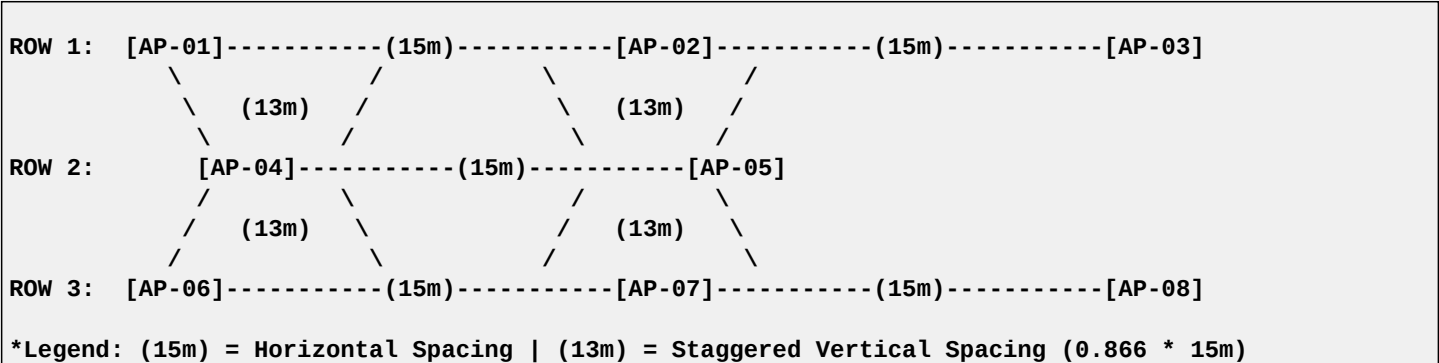


REFINED EXPERT RF DESIGN: TRIANGULAR STAGGERED LAYOUT

1. VISUAL DESIGN LAYOUT (STAGGERED TRIANGLE)

The staggered triangular layout ensures that the 'Weak Spot' of any two APs is covered directly by a third AP. This eliminates coverage holes.



2. DISTANCE & SPACING FORMULAS

- To achieve expert-level mobilization for Voice, the 'Cell Overlap' must be at least 20%.
- For 5GHz (Standard Enterprise): Distance between APs should be 15 meters (approx 50 feet).
 - For High Ceiling / Industrial: Distance should be reduced to 12 meters (approx 40 feet).

Why this distance?

In an indoor environment with standard walls, a Wi-Fi signal drops to -65 dBm (the threshold for voice) at approximately 7-8 meters from the AP. By placing APs 15 meters apart, the 'overlap' occurs at the 7.5-meter mark, ensuring continuous -65 dBm coverage throughout the floor.

3. CHANNEL REUSE STRATEGY

- In a triangular grid, every adjacent AP must use a non-overlapping channel to prevent Co-Channel Interference (CCI).
- 2.4 GHz Plan: AP-01 (Ch 1), AP-02 (Ch 6), AP-04 (Ch 11).
 - 5 GHz Plan: AP-01 (Ch 36), AP-02 (Ch 44), AP-04 (Ch 52).
- This creates a 'Channel Cluster' where no two APs in a triangle share the same frequency.

4. SUMMARY FOR ARCHITECTS

1. Design for 5GHz: Always plan based on 5GHz propagation; 2.4GHz will naturally follow but must be power-tuned (reduced) to match 5GHz cells.
2. Fast Roaming: Enable 802.11r (Fast Transition) to allow the 'Mobilization' required for Voice.
3. Triangular Spacing: Use the 15-meter staggered rule to ensure N+1 redundancy.